Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details:

Batch Number: 13

Start date: 02-02-2026

Name: S Sneha

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX AI-Driven Urban Mobility Optimization System

2. Project Objective

The primary objective of this project is to design and develop an intelligent urban mobility system that reduces traffic congestion and improves fleet efficiency. NeuroFleetX aims to optimize vehicle routing, predict maintenance needs, and enhance user experience through AI-driven recommendations. A key performance indicator (KPI) considered in this project is **a**verage travel time per trip, which reflects the system's effectiveness in traffic optimization.

3. Project description in detail

NeuroFleetX is an AI-powered platform designed to enhance urban mobility by integrating real-time data, machine learning, and intelligent decision-making systems.

The system includes multiple modules such as authentication, fleet tracking, route optimization, predictive maintenance, and customer booking. The backend is developed using Spring Boot, while the frontend uses React/Angular for interactive dashboards. AI functionalities are implemented using Python-based microservices, and data is managed using **MySQL.**

The platform uses simulated IoT data for vehicle telemetry, including location, speed, and fuel/battery levels. AI-based routing algorithms (such as Dijkstra combined with predictive models) help determine optimal routes considering traffic conditions. Predictive maintenance alerts are generated based on vehicle health parameters.

In real-world implementation, NeuroFleetX can be used in smart cities, ride-sharing systems, logistics, and EV fleet management. It helps reduce congestion, improve efficiency, and promote sustainable mobility solutions.

4. Timeline Overview

Week	Activities Planned	Activities Completed

Week 1	Project understanding, requirement analysis	Finalized project idea and modules
Week 2	UI design and authentication setup	Designed login/register screens
Week 3	Backend setup and database design	Implemented APIs and schema
Week 4	Fleet module development	Vehicle CRUD and telemetry simulation
Week 5	Route optimization integration	Map APIs and route logic implemented
Week 6	Predictive maintenance module	Health analytics dashboard created
Week 7	Booking system and recommendations	Customer UI and AI logic completed
Week 8	Testing, optimization, documentation	Final deployment and report prepared

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial planning and team alignment	Week 1
Prototype/First Draft	Basic UI and backend setup	Week 3
Mid-Term Review	Core modules completed	Week 5
Final Submission	Fully integrated system	Week 8
Presentation	Final demo and explanation	Week 8

5b. Project execution details

Milestone 1: 12-Feb-2026

I have successfully completed the development of the Login and Sign-Up pages for the NeuroFleetX project. I implemented these features by applying my knowledge of React for the frontend and Spring Boot for the backend integration. The authentication flow has been structured to ensure proper communication between the client and server. Additionally, I strengthened my backend fundamentals by practicing Java concepts. I also enhanced my database skills by working with MySQL for data handling and storage.

Milestone 2: 27-Feb-2026

I have successfully completed the development of the Fleet Inventory, AI Route Optimization, and Load Balancing modules for the NeuroFleetX project. The Fleet Inventory system was designed to efficiently manage vehicle details, availability status, and resource allocation using a structured database approach. I implemented AI-based Route Optimization to determine the most efficient travel paths by analyzing factors such as distance, traffic conditions, and delivery priorities, improving operational efficiency. Additionally, the Load Balancing module was developed to distribute tasks and shipments evenly across available fleet resources, ensuring optimal utilization and preventing overload on individual vehicles. These features were built using React for the frontend, Spring Boot for backend services, and MySQL for reliable data storage, further strengthening my skills in full-stack development, system design, and performance optimization.

Milestone 3: 06-Mar-2026

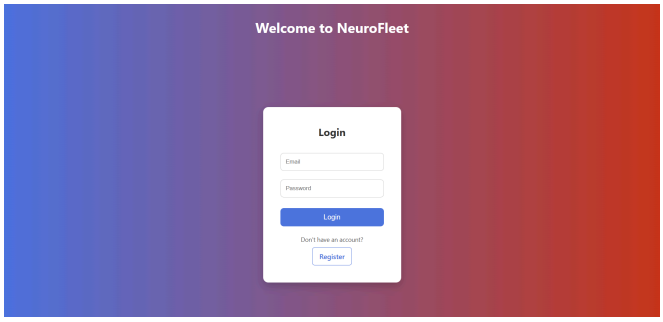
For Milestone 3 of the project, two key modules were developed: Predictive Maintenance & Health Analytics and Customer Booking & Smart Recommendations. The Predictive Maintenance & Health Analytics module focuses on monitoring system or vehicle health by analyzing operational data and identifying potential issues before failures occur. By leveraging analytics techniques, the module helps predict maintenance requirements, track performance trends, and provide early alerts, thereby reducing downtime and improving reliability. The Customer Booking & Smart Recommendations module enables users to easily schedule services through an intuitive booking interface. In addition to basic booking functionality, the system provides intelligent recommendations based on user preferences, past service history, and availability. This enhances the user experience by suggesting optimal service times or relevant maintenance services. Together, these modules improve operational efficiency, enable proactive maintenance, and deliver a more personalized and convenient experience for customers.

Milestone 4: 20-Mar-2026

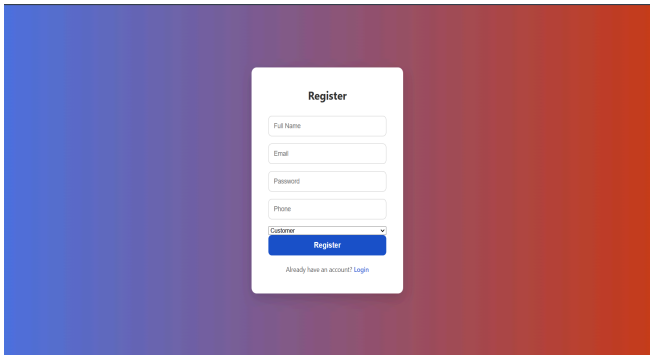
For Milestone 4 of the project, two important modules were successfully developed: Admin Dashboard and Urban Mobility Insights. The Admin Dashboard serves as a centralized interface that allows administrators to monitor, manage, and control various system operations efficiently. It provides a clear overview of key metrics such as user activity, bookings, fleet status, and system performance, enabling better decision-making and streamlined management. The Urban Mobility Insights module focuses on analyzing transportation and usage data to derive meaningful patterns and trends. By leveraging data analytics, this module helps in understanding traffic flow, demand distribution, and operational efficiency across different regions. These insights support strategic planning and optimization of resources, ultimately contributing to improved mobility solutions and enhanced overall system performance.

6. Snapshots / Screenshots

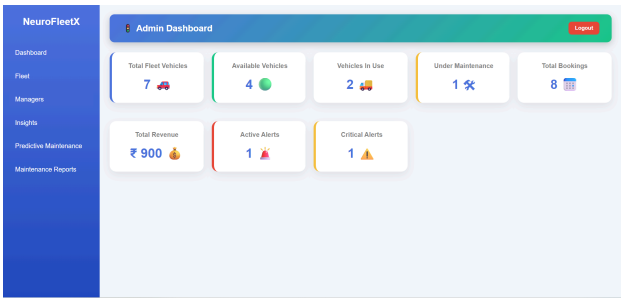
Login UI



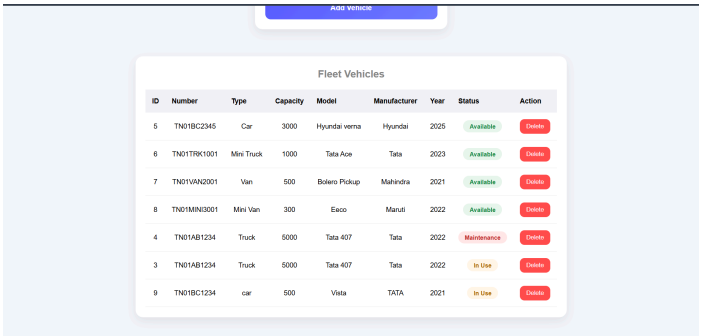
Register UI



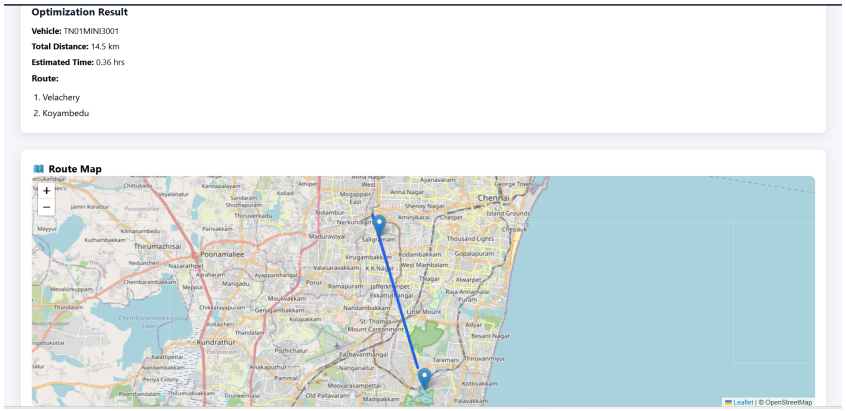
Dashboard



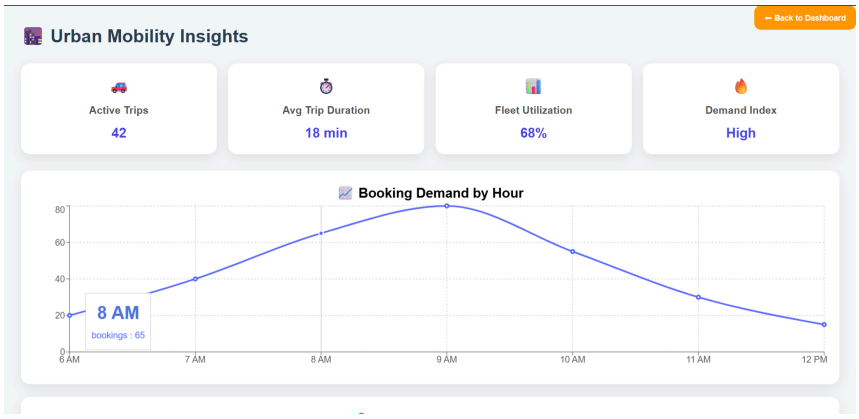
Fleet cards



Map with routes



Analytics charts



7. Challenges Faced

During the project, several challenges were encountered:

Integration complexity: Combining frontend, backend, and AI modules required careful coordination.

API handling: Working with map APIs and real-time simulation was initially difficult.

Data simulation: Creating realistic vehicle telemetry data required multiple iterations.

Team coordination: Managing tasks across multiple modules needed proper planning.

These challenges were resolved through team discussions, research, and iterative testing.

8. Learnings & Skills Acquired

This internship helped in gaining both technical and soft skills:

Technical Skills:

Spring Boot & REST APIs

React/Angular frontend development

Database design using MySQL

AI concepts and predictive analytics

API integration (Maps, WebSockets)

Soft Skills:

Team collaboration

Problem-solving

Time management

Project planning

9. Testimonials from team

Working on NeuroFleetX was a highly enriching experience. The project helped us understand real-world applications of AI in smart mobility. We learned how to work as a team, divide responsibilities, and deliver a complete product within deadlines. The internship improved our confidence and technical abilities significantly.

10. Conclusion

The NeuroFleetX project successfully demonstrates how AI can be used to optimize urban mobility systems. It provides solutions for reducing traffic congestion, improving fleet efficiency, and enhancing user experience.

This internship aligned well with our academic learning and provided practical exposure to real-world technologies. It has strengthened our interest in AI, full-stack development, and smart city solutions.

11. Acknowledgements

We would like to express our sincere gratitude to Infosys Springboard for providing this valuable internship opportunity. We also thank our mentor for their continuous guidance and support. Lastly, we appreciate our team members for their dedication and collaboration throughout the project.